RW–070

Sub. Code
511201

M.Sc. DEGREE EXAMINATION, APRIL 2017

Second Semester

Mathematics

ALGEBRA — II

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define a vector space.

2. Define homomorphism

3. Define an algebraic number.

4. Define splitting field.

5. When will K be a normal extension of field F?

6. Define Galois group of $f(x)$.

7. Define right invertible.

8. Define index of nilpotence.

9. Define trace.

10. Define hermitian adjoint.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) If V is finite-dimensional over F then prove that any two bases of V have the same number of elements.

Or

(b) If V is the internal direct sum of $U_1 \ldots \ldots U_n$ then show that V is isomorphic to the external direct sum of $U_1 \ldots\ldots U_n$.

12. (a) If F is of characteristic 0 and if a, b are algebraic over F, then prove that there exists an element $c \in F(a, b)$ such that $F(a, b) = F(c)$.

Or

(b) Prove that a polynomial of degree n over a field can have atmost n roots in any extension field.

13. (a) Show that K is a normal extension of F if and only if K is the splitting field of some polynomial over F.

Or

(b) Prove that S_n is not solvable for $n \geq 5$.

14. (a) Prove that if V is finite-dimensional over F, then $T \in A(V)$ is regular if and only if T maps V onto V.

Or

(b) Show that any two nilpotent linear transformations are similar if and only if they have the same invariants.

15. (a) Prove that the linear transformation T on V is unitary if and only if it takes an orthonormal basis of V into an orthonormal basis of V.

Or

(b) If $T \in A(V)$ is hermitian, then prove that all its characteristic roots are real.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Prove that if V is finite dimensional and if W is a subspace of V, then W is finite-dimensional, $\dim W \leq \dim V$ and $\dim V/W = \dim V - \dim W$.

17. Show that the element $a \in K$ is algebraic over F if and only if $F(a)$ is a finite extension of F.

18. If K is a finite extension of F, then $G(K, F)$ is a finite group and its order, $o(G(K, F))$ satisfies $o(G(K,F)) \leq [K, F]$.

19. If $T \in A(V)$ has all its characteristic roots in F, then prove that there is a basis of V in which the matrix of T is triangular.

20. If N is a normal linear transformation on V, then prove that there exists an orthonormal basis, consisting of characteristic vectors of N, in which the matrix of N is diagonal. Equivalently, if N is a normal matrix then there exists a unitary matrix U such that UNU^{-1} $(= UNU^*)$ is diagonal.

RW–071

Sub. Code
511202

M.Sc. DEGREE EXAMINATION, APRIL 2017

Second Semester

Mathematics

MEASURE AND INTEGRATION

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State monotonicity property.

2. Define measurable function.

3. Define simple function of the measurable sets E_i.

4. When we say a measurable function is integrable.

5. Define absolutely continuous function.

6. Define Lebesgue integrals.

7. When we say a set is measurable?

8. State Hahn Decomposition Theorem.

9. Define outer measure μ^* of a set.

10. Define product measure.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) If E_1 and E_2 are measurable, then prove that $E_1 \cup E_2$ is also measurable.

Or

(b) Prove that every Borel set is measurable.

12. (a) Let φ and ψ be simple functions which vanish outside a set of finite measure. Then prove that $\int (a\varphi + b\psi) = a\int \varphi + b\int \psi$ and if $\varphi \geq \psi$ a.e., then prove that $\int \varphi \geq \int \psi$.

Or

(b) State and prove Fatou's lemma.

13. (a) If f is absolutely continuous on $[a, b]$, then show that it is of bounded variation on $[a, b]$.

Or

(b) State and prove Lebesgue Convergence theorem.

14. (a) State and prove Fubini's theorem.

Or

(b) State and prove Monotone Convergence theorem.

15. (a) Prove that if $A \in \mathfrak{A}$, then A is measurable with respect to μ^*.

Or

(b) Let $\{(A_i \times B_i)\}$ be a countable disjoint collection of measurable rectangles whose union is a measurable rectangle $A \times B$. Then prove that $\lambda(A \times B) = \sum \lambda(A_i \times B_i)$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Prove that the outer measure of an interval is its length.

17. If f and g are bounded measurable functions defined on a set E of finite measure, then Prove that following :

    (a) $\int_E (af + bg) = a\int_E f + b\int_E g$

    (b) If $f = g$ a.e., then $\int_E f = \int_E g$

    (c) If $f \leq g$ a.e., then $\int_E f \leq \int_E g$

    (d) If $A \leq f(x) \leq B$, then $AmE \leq \int_E f \leq BmE$

    (e) If A and B are disjoint measurable sets of finite measure, then $\int_{A \cup B} f = \int_A f + \int_B f$.

18. State and prove Vitali's lemma.

19. State and prove Radon-Nikodym theorem.

20. Prove that the class $\mathfrak{B}$ of μ^*-measurable sets is a σ-algebra. If $\overline{\mu}$ is μ^* restricted to $\mathfrak{B}$, then $\overline{\mu}$ is a complete measure on $\mathfrak{B}$.

RW–072

Sub. Code
511203

M.Sc. DEGREE EXAMINATION, APRIL 2017

Second Semester

Mathematics

PARTIAL DIFFERENTIAL EQUATION

(CBCS 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write down the Pfaffian differential equation.

2. Verify whether the equation $yz(y+z)dx + xz(x+z)dy + xy(x+y)dz = 0$ is integrable.

3. Eliminate the constants a and b from the following equation : $2z = (ax+y)^2 + b$.

4. Define plane element.

5. Show that the equations $xp = yq, z(xp+yq) = 2xy$ are compatible.

6. Find the complete integral of the equation $z^2 = pq\,xy$.

7. Write the telegraphy equation.

8. Find the particular integral of the equation $(D^2 - D')z = e^{2x+y}$.

9. Explain how the magnetostatics field equations be reduced to Laplace's equation.

10. Write the formula for kinematic viscosity.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Find the integral curves of the equations.

$$\frac{dx}{y(x+y)+az} = \frac{dy}{x(x+y)-az} = \frac{dz}{z(x+y)}.$$

Or

(b) Find the orthogonal trajectories on the conicoid $(x+y)z = 1$ of the conics in which it is cut by the system of planes $x - y + z = k$ where k is a parameter.

12. (a) Find the equation of the system of surfaces which cut orthogonally the cones of the system $x^2 + y^2 + z^2 = cxy$.

Or

(b) Prove that the function $F(x, y, z, p, q)$ is a constant along every characteristics strip of the equation $F(x, y, z, p, q) = 0$.

13. (a) Find a complete integral of the partial differential equation.

$(p^2 + q^2)x = pz$ and deduce the solution which passes through the curve $x = 0, z^2 = 4y$.

Or

(b) Show that the only integral surface of the equation $2q(z - px - qy) = 1 + q^2$ which is circumscribed about the paraboloid $2x = y^2 + z^2$ is the enveloping cylinder which touches it along its section by the plane $y + 1 = 0$.

14. (a) Show that the equation $\frac{\partial^2 z}{\partial x^2} = \frac{1}{k}\frac{\partial z}{\partial t}$ possesses solutions of the form $\sum_{n=0}^{\infty} c_n \cos(nx + \in_n) e^{-kn^2 t}$.

Or

(b) Solve the equation $\frac{\partial^4 z}{\partial x^4} + \frac{\partial^4 z}{\partial y^4} = 2\frac{\partial^4 z}{\partial x^2 \partial y^2}$.

15. (a) Explain briefly about the elementary solutions of the one-dimensional wave equation.

Or

(b) Explain the conduction of heat by the theory of diffusion equation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find the orthogonal trajectories of a system of plane curves on a surface.

17. Find the general integral of the equation $(x - y)p + (y - x - z)q = z$ and the particular solution through the circle $z = 1, x^2 + y^2 = 1$.

18. Find the complete integral of the following equations by Charpit's method

(a) $px^5 - 4q^3x^2 + 6x^2z - 2 = 0$

(b) $2(z + xp + yq) = yp^2$.

19. Determine the solution of the equation $\frac{\partial^4 z}{\partial x^4} + \frac{\partial^2 z}{\partial y^2} = 0$ $(-\infty < x < \infty,\ y \geq 0)$ satisfying the conditions :

(a) z and its partial derivatives tend to zero as $x \to \pm\infty$;

(b) $z = f(x),\ \partial z / \partial y = 0$ on $y = 0$.

20. Discuss briefly about the elementary solutions of the diffusion equation.

RW–073

Sub. Code
511204

M.Sc. DEGREE EXAMINATION, APRIL 2017

Second Semester

Mathematics

COMPLEX ANALYSIS

(CBCS 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Show that a harmonic function satisfies the formal differential equation $\frac{\partial^2 u}{\partial z \partial \bar{z}} = 0$.

2. Find the linear transformation which carries 0, i, $-i$ into 1, –1, 0.

3. Evaluate $\int_C \frac{dz}{z-a}$ where C is the circle with centre a.

4. Compute $\int_{|z|=1} e^z z^{-n} dz$.

5. Show that the function e^z is essential singularities at ∞.

6. If $f(z)$ is analytic at z_0 with $f'(z_0) \neq 0$, prove that it maps a neighborhood of z_0 conformally and topologically onto a region.

7. How many roots of the equation $z^4 - 6z + 3 = 0$ have their modulus between 1 and 2?

8. Find the poles and residues of the function $\frac{1}{z^2+5z+6}$.

9. If the function $f_n(z)$ are analytic and $\neq 0$ in a region Ω and if $f_n(z)$ converges to $f(z)$ uniformly on every compact subset of Ω, prove that $f(z)$ is either identically zero or never equal to zero in Ω.

10. Show that $\prod_{n=2}^{\infty}\left(1-\frac{1}{n^2}\right)=\frac{1}{2}$.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) If all zeros of a polynomial $P(z)$ lie in a half plane, prove that all zeros of the derivative $P'(z)$ lie in the same half plane.

Or

(b) Show that any linear transformation which transforms the real axis into itself can be written with real coefficients.

12. (a) Prove that the line integral $\int_\gamma pdx+qdy$ defined in Ω, depends only on the end points of γ if and only if there exists a function $U(x,y)$ in Ω with the partial derivatives $\partial U/\partial x = p$, $\partial U/\partial y = q$.

Or

(b) Suppose that $\varphi(\zeta)$ is continuous on the arc γ. Prove that the function $F_n(z)=\int_\gamma \frac{\varphi(\zeta)d\zeta}{(\zeta-z)^n}$ is analytic in each of the regions determined by γ and its derivative is $F_n'(z)=nF_{n+1}(z)$.

13. (a) Suppose that $f(z)$ is analytic in the region Ω' obtained by omitting a point a from a region Ω. Prove that a necessary and sufficient condition that there exist an analytic function in Ω which coincides with $f(z)$ in Ω' is that $\lim_{z \to a} (z-a)f(z) = 0$. The extended function is uniquely determined.

Or

(b) If $f(z)$ is analytic and non-constant in a region Ω, prove that its absolute value $|f(z)|$ has no maximum in Ω.

14. (a) If $f(z)$ is analytic in a region Ω, prove that $n(\gamma, a)f(a) = \frac{1}{2\pi i}\int_{\gamma} \frac{f(z)dz}{z-a}$ for every cycle γ which is homologous to zero in Ω.

Or

(b) Let γ be homologous to zero in Ω and such that $n(\gamma, z)$ is either 0 or 1 for any point z not on γ. Suppose that $f(z)$ and $g(z)$ are analytic in Ω and satisfy the inequality $|f(z) - g(z)| < |f(z)|$ on γ. Prove that $f(z)$ and $g(z)$ have the same number of zeros enclosed by γ.

15. (a) Show that the Laurent development of $(e^z - 1)^{-1}$ at the origin is of the form $\frac{1}{z} - \frac{1}{2} + \sum_{1}^{\infty} (-1)^{k-1} \frac{B_k}{(2k)!} z^{2k-1}$ where the numbers B_k are known as the Bernoulli numbers. Calculate B_1, B_2, B_3.

Or

(b) Prove that a necessary and sufficient condition for the absolute convergence of the product $\prod_{1}^{\infty} (1 + a_n)$ is the convergence of the series $\sum_{1}^{\infty} |a_n|$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If $u(x,y)$ and $v(x,y)$ have continuous first-order partial derivatives which satisfy the Cauchy-Riemann differential equations, prove that $f(z)=u(z)+iv(z)$ is analytic with continuous derivative $f'(z)$ and conversly.

17. If $f(z)$ is analytic in an open disk Δ, prove that $\int_\gamma f(z)dz = 0$ for every closed curve γ in Δ.

18. Prove that if $f(z)$ is analytic in a region Ω, containing a, it is possible to write $f(z)=f(a)+\frac{f'(a)}{1!}(z-a)+\frac{f''(a)}{2!}(z-a)^2+\ldots+\frac{f^{(n-1)}(a)}{(n-1)!}(z-a)^{n-1}+f_n(z)(z-a)^n$, where $f_n(z)$ is analytic in Ω.

19. If $f(z)$ is meromorphic in Ω with the zeros a_j and the poles b_k, prove that $\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}dz = \sum_j n(\gamma,a_j)-\sum_k n(\gamma,b_k)$ for every cycle γ which is homologous to zero in Ω and does not pass through any of the zeroes or poles.

20. Let $\{b_v\}$ be a sequence of complex numbers with $\lim_{v\to\infty} b_v = \infty$ and let $P_v(\zeta)$ be polynomials without constant term. Prove that there are functions which are meromorphic in the whole plane with poles at the points b_v and the corresponding singular parts $P_v(1/(z-b_v))$. Moreover, the most general meromorphic function of this kind can be written in the form $f(z)=\sum_v\left[P_v\left(\frac{1}{z-b_v}\right)-p_v(z)\right]+g(z)$ where the $p_v(z)$ are suitably chosen polynomials and $g(z)$ is analytic in the whole plane.

RW–075

Sub. Code
511401

M.Sc. (Maths) DEGREE EXAMINATION, APRIL 2017

Fourth Semester

FUNCTIONAL ANALYSIS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Give an example of a closed and bounded of a metric space which is not compact.

2. Define a bounded linear operator.

3. If $x \perp y$ in an inner product space, show that $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.

4. Show that any orthonormal set is linearly independent.

5. If $\langle v_1, w\rangle = \langle v_2, w\rangle$ for all w in an inner product space X, prove that $v_1 = v_2$.

6. If S and T are bounded linear operators, prove that $(S+T)^* = S^* + T^*$.

7. Define a sublinear functional.

8. If a normed space X is reflexive, prove that it is complete.

9. Show that the weak limit X of (x_n) is unique.

10. Prove that $\|(x,y)\| = \|x\| + \|y\|$ defines a noun on $X \times Y$. Where X and Y are normed spaces.

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) On a finite dimensional vector space X prove that any norm $\|\cdot\|$ is equivalent to any other norm $\|\cdot\|_0$.

Or

(b) Let $T : D(T) \to y$ be a linear operator, where $D(T) \subset X$ and X, Y are normal spaces. Prove that T is continuous if and only if T is bounded.

12. (a) State and prove schwary inequality.

Or

(b) Let H be a Hilbert space. Prove that if H is separable, then every orthonormal set in H is countable.

13. (a) Let H_1, H_2 be Hilbert spaces and $T : H_1 \to H_2$ bounded linear operator. Prove that $(T^*)^* = T$ and $\|T^* T\| = \|T T^*\| = \|T\|^2$.

Or

(b) Let $T : H \to H$ be a bounded linear operator on a Hilbert space H. Prove that

(i) T is self – adjoint $\Rightarrow \langle T_x, x \rangle$ is real for all $x \in H$.

(ii) If H is complex and $\langle T_x, x \rangle$ is real for all $x \in H$, then the operator T is self adjoint.

14. (a) Let $T : X \to Y$ be a bounded linear operator where X and Y are normed spaces. Prove that the adjoint operator T^x is linear and bounded and $\|T^x\| = \|T\|$.

Or

(b) Show that the normed space X_n itself is separable if the dual space X' of X is separable.

15. (a) Let (x_n) be a weakly convergent sequence in a normed space X prove that the sequence $(\|x_n\|)$ is bounded.

Or

(b) State and prove a criterion for strong operator convergence.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. If Y is a Banach space, prove that $B(x, y)$ is a Banach space.

17. Let X be an inner product space and $m \neq \phi$ a convex subset which is complete. Then for every given $x \in X$, prove that there exists a unique $y \in m$ such that $\delta = \inf_{\bar{y} \in m} \|x - \bar{y}\| = \|x - y\|$.

18. State and prove Riesz's theorem.

19. State and prove Hahn – banach theorem for. Real vector spaces.

20. Define a closed linear operator. State and prove the closed graph theorem.

RW–076

Sub. Code
511402

M.Sc. DEGREE EXAMINATION, APRIL 2017.

Fourth Semester

Mathematics

PROBABILITY AND STATISTICS

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. If the sample space is $\mathcal{C} = C_1 \cup C_2$ and if $P(C_1) = 0.8$ and $P(C_2) = 0.5$, find $P(C_1 \cap C_2)$.

2. Let x have the p.d.f. $f(x) = \begin{cases} 4x^3, & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$ find $E(x)$.

3. Let x_1 and x_2 have the point p.d.f. of

   $f(x_1, x_2) = \begin{cases} 6x_2 & 0 < x_2 < x_1 < 1 \\ 0 & \text{elsewhere} \end{cases}$ find the marginal p.d.f. of x_1.

4. Define the covariance and correlation coefficient of two random variably.

5. If the m.g.f of a random variable x is $\left(\frac{1}{3} + \frac{2}{3}et\right)^5$, find σ^2.

6. Define the gamma distribution.

7. If $x_1 = y_1 y_2$ and $x_2 = y_1 - y_1 y_2$, find the Jacobian J.

8. Define the "F" distribution.

9. Let W_n denote a random variable with mean μ and variance $\frac{b}{n^p}$ where $p > 0$, μ and b are constants. Prove that W_n converges stochastically to μ.

10. Define convergence in probability.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) If C_1 and C_2 are subsets of $\mathcal{C}$, prove that $P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$ and $P(\phi) = 0$.

Or

(b) Let $f(x) = \frac{1}{2}, -1 < x < 1$, zero elsewhere, lets p.d.f. of the random variable x. Define y by $y = x^2$. Find the p.d.f of y.

12. (a) Let x_1 and x_2 have the point p.d.f.

$f(x_1, x_2) = \begin{cases} 2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$ find the marginal density functions and the conditional p.d.f. of x_1, given $x_2 = x_2$.

Or

(b) Prove that $E[E(x_2 / x_1)] = E(x_2)$.

13. (a) Find the m.g.f of a binomial distribution and hence find mean and the variance.

Or

(b) If the random variable x has a Poisson distribution such that $P_n(x=1)=P_n(x=2)$, find $P_n(x=4)$.

14. (a) Let x have the p.d.f. $f(x)=\left(\frac{1}{2}\right)^x$, $x=1,2,3,....$ and zero elsewhere find the p.d.f of $y=x^3$.

Or

(b) Let x_1 and x_2 be independent with normal distributions $N(\mu_1 \sigma_1^2)$ and $N(\mu_2,\sigma_2^2)$ respectively. Define the random variable y by $y=x_1-x_2$. Find the p.d.f. of y.

15. (a) Let x_i denote a random variable with mean μ_i and variable $\sigma_i^2, i=1,2,...n$. Let $x_1,x_2,....x_n$ be independent and let $k_1,k_2...,$ k_n denote and constants. Let $y=\sum_{i=1}^{n} k_i x_i$. Compute the mean and variance of y.

Or

(b) Let U_n and V_n converge in probability to c and d respectively. Prove that the sum U_n+V_n converges in probability to $c+d$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find the moments of the distribution that has m.g.f $m(t) = (1-t)^{-3}\ t < 1$.

17. Let the random variable x and y have the joint p.d.f $f(x,y) = x + y, 0 < x < 1, 0 < y < 1$, zero elsewhere. Compute the correlation coefficient of x and y.

18. If the random variable x is $N(\mu, \sigma^2), \sigma^2 > 0$, prove that the random variable $V = (x, \mu)^2 / \sigma^2$ is $\chi^2(1)$.

19. Let x_1, x_2, x_3 denote a random sample of size 3 from a distribution that is $N(0,1)$ find the p.d.f of $y = x_1^2 + x_2^2 + x_3^2$.

20. State and prove the central limit theorem.

RW–077

Sub. Code
511502

M.Sc. DEGREE EXAMINATION,
APRIL 2017

Fourth Semester

Mathematics

***Elective* IV — GRAPH THEORY**

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Show that $\delta \le 2\frac{\varepsilon}{\gamma} \le \Delta$.

2. Draw any two trees on six vertices.

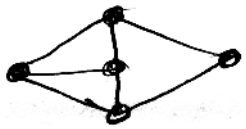
3. Define the connectivity $K(G)$ of a graph G with an example.

4. If G is hamiltonian, prove that for every non empty proper subset S of V, $W(G-S) \le |S|$.

5. Define a maximum matching with an example.

6. For the following graph G find α and β.

7. Find the edge chromatic number of the following graph



8. Find $\pi_k(K_n)$.

9. If a graph G is embeddable on the sphere, prove that it is embeddable in the plane.

10. If G is a plane graph, prove that

$$\sum_{f \in F} d(f) = 2\varepsilon$$

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Show that $\sum_{v \in V} d(v) = 2\varepsilon$ and hence prove that, in any graph, the number of vertices of odd degree is even.

Or

(b) If G is a tree, prove that $\varepsilon = r - 1$.

12. (a) With usual notations, prove that $K \le K' \le \delta$.

Or

(b) Define the closure $C(G)$ of a graph G and show that $C(G)$ is well defined.

13. (a) Define a maximum matching. If m is a maximum matching of G, prove that G contains no m-augmenting path.

Or

(b) If G is simple and contains no k_{m+1}, prove that $\varepsilon(G) \leq \varepsilon(T_m, r)$. Also prove that $\varepsilon(G) = \varepsilon(T_m, r)$ only if $G \cong T_m, r$.

14. (a) Let G be a connected graph that is not an odd cycle. Prove that G has a 2-edge colouring in which both colours are represented at each vertex of degree at least two.

Or

(b) If G is k - critical, prove that $\delta \geq k-1$.

15. (a) Prove that K_5 is non planar.

Or

(b) State and prove Euler's formula.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let G be a graph with $\gamma - 1$ edges. Show that the following three statements are equivalent

(a) G is connected

(b) G is a cyclic

(c) G is a tree.

17. State and prove Whitney's theorem.

18. Prove that G has a perfect matching if and only if $O(G-S) \leq |S|$ for all $S \subseteq V$.

19. State and prove Vizing's theorem.

20. Prove that every planar graph is 5-vertex colourable.

RW–078

Sub. Code
511504

M.Sc. (Maths) DEGREE EXAMINATION, APRIL 2017

Fourth Semester

Elective V — OPTIMIZATION TECHNIQUES

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the difference between a critical and a noncritical activity?

2. For an activity (i, j) define $\square_i$ and Δ_j.

3. Show that $Q = \{(x_1, x_2) / x_1 + x_2 \le 1,\ x_1 \ge 0, x_2 \ge 0\}$ is convex.

4. State any two advantages of the revised simplex method over the regular simplex method.

5. Define a strictly determinable game with an example.

6. Find the minimax and maximin value of the game $\begin{bmatrix} 1 & 3 & 6 \\ 2 & 1 & 3 \\ 6 & 2 & 1 \end{bmatrix}$.

7. Define an extreme point of a function.

8. Define the Lagrangean function and the Lagrange multipliers.

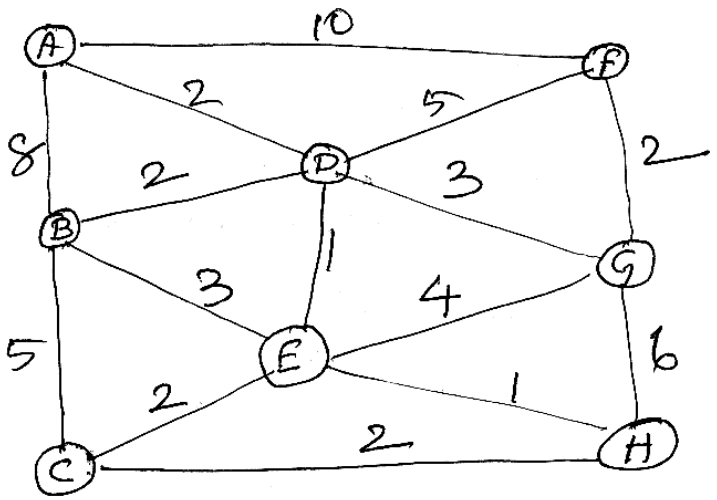
9. Explain Steepest ascent method.

10. Give an example for a nonseparable programming.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Solve the minimum-span problem for the network given below.



Or

(b) Write down the steps of the maximal flow algorithm.

12. (a) Determine and classify all the basic solutions of the following system of equations.

$$\begin{pmatrix} 5 & 4 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Or

(b) Write down the steps of the revised simplex algorithm.

13. (a) Solve the following game whose pay-off matrix is given by

Player B

		B_1	B_2	B_3	B_4	B_5
	A_1	9	3	1	8	0
Player A	A_2	6	5	4	6	7
	A_3	2	4	3	3	8
	A_4	5	6	2	2	1

Or

(b) Solve the following 2×3 game graphically.

Player B

		B_1	B_2	B_3
Player A	A_1	1	3	11
	A_2	8	5	2

14. (a) Obtain a sufficient condition for a stationary point X_0 to be an extremum.

Or

(b) Examine the following function for extreme points $f(X) = x_1 + 2x_3 + x_2x_3 - x_1^2 - x_2^2 - x_3^2$.

15. (a) Maximize $f(x) = \begin{cases} 3x, & 0 \le x \le 2 \\ \frac{1}{3}(-x+20), & 2 \le x \le 3. \end{cases}$

Or

(b) Show how the following problem can be made seperable.

Min $z = e^{2x_1 + x_2^2} + (x_3 - 2)^2$

Subject to

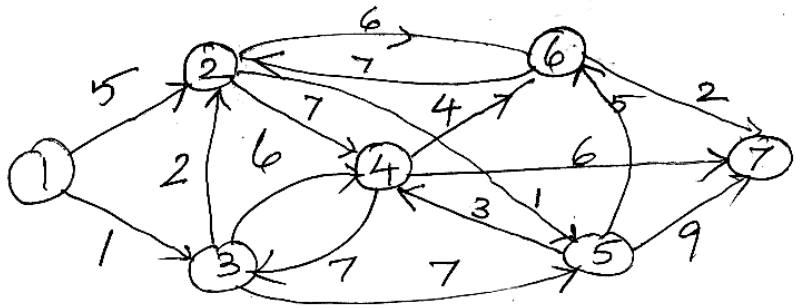
$x_1 + x_2 + x_3 \le 6$

$x_1, x_2, x_3 \ge 0.$

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Use Dijkstra's algorithm to find the shortest route between node 1 and every other node in the network.



17. Use revised simplex method to solve the following LPP.

Min $z = x_1 + 2x_2$

Subject to

$2x_1 + 5x_2 \geq 16$

$x_1 + x_2 \geq 2$

$x_1, x_2 \geq 0.$

18. Solve the following game of using simplex method.

	Player B		
Player A	1	–1	3
	3	5	–3
	6	2	–2

19. Minimize $f(x) = x_1^2 + x_2^2 + x_3^2$

Subject to

$g_1(x) = x_1 + x_2 + 3x_3 - 2 = 0$

$g_2(x) = 5x_1 + 2x_2 + x_3 - 5 = 0.$

20. Maximize $f(x_1, x_2) = 4x_1 + 6x_2 - 2x_1^2 - 2x_1x_2 - 2x_2^2$ by the steepest-ascent method.